

Empirical Proof Record: REFUTED

2026-05-23T20:32:21.688210+00:00

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Proof ID: 232db71cba37509efbef3d2de9938c08...

Created: 2026-05-23T20:32:21.688210+00:00

1. Claim

At least 80% of the always-on numeric signals Kimera emits (those present in EVERY cycle) carry real dynamics (vary with the stimulus) rather than emitting a frozen default constant.

- **Operationalisation:** Stream 64 diverse records from the 'flores' corpus through the Kimera 'entity' target; collect every finite numeric leaf in CycleResult.raw per cycle. Restrict to ALWAYS-ON signals (present in all cycles $\rightarrow$ zero sampling doubt); a signal is LIVE if it takes ≥ 2 distinct values, FROZEN if constant. $\text{liveness_rate} = \text{always_on_live} / \text{always_on_total}$. Under-sampled signals are excluded (a signal seen in few cycles is trivially constant — not a frozen default).
- **Threshold:** $\text{liveness_rate} \geq 0.8$ proportion
- **H0:** H0: $\text{liveness} < 80\%$ — most emitted signals are frozen defaults; the body is structurally wired but not dynamically alive; the frozen worklist is non-empty
- **H1:** H1: $\text{liveness} \geq 80\%$ — the emitted signal surface is dynamically alive

2. Verdict

REFUTED — observed 0.6715506715506715

always-on liveness 0.6716: 550/819 every-cycle signals carry real dynamics; 269 are frozen defaults (the worklist); Wilson 95% CI [0.6387, 0.7028]. Context: whole-surface 3311/15754 (0.2102, deflated by under-sampling); 64 cycles

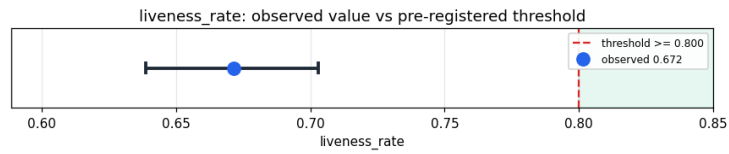


Figure 1: liveness_rate – observed value with Wilson 95% CI.

3. Evidence

Pillar	Statistic	Value	95% CI	Library
signal_dynamics	liveness_rate	0.6715506715506715	(0.6387, 0.7028)	statsmodels 0.14.6

4. Pre-registration

- Registered at: 2026-05-23T20:30:07.782503+00:00
- Config hash: e683f31a2ae22e752e0db448f4a30594...
- Data hash: bf4196403365897ad95dbcc18fc5116a...

5. Signature

858764d965bfaaadf36377a56493caf88825d3712cdacd629a6dbefa049ab8e9